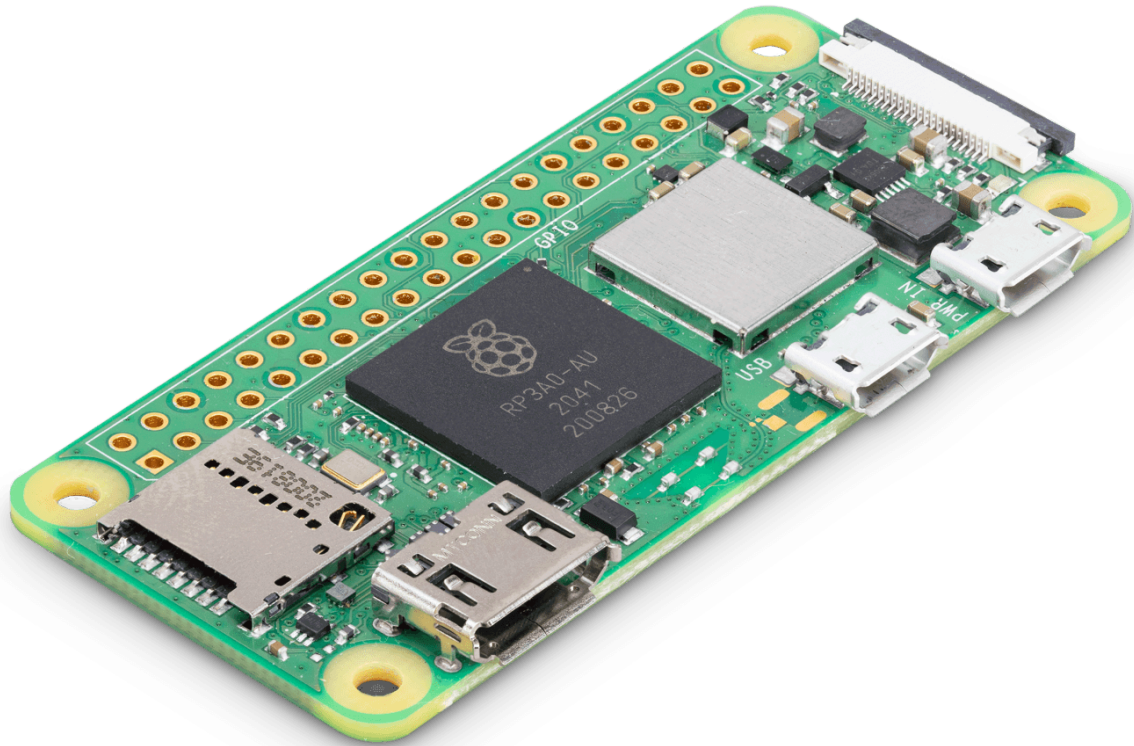


Lab 001 - Initial Raspberry Pi Deployment



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Completed 6/30/26

Comprehensive Lab Report

Objective

Deploy a Raspberry Pi Zero W running Raspberry Pi OS and configure secure remote administration over SSH.

Hardware & Materials

- Raspberry Pi Zero W
- 128 GB MicroSD Card
- Windows 11 PC/Ubuntu Server
- Home Wi-Fi

Tasks Completed

- Installed Raspberry Pi OS
- Configured Wi-Fi
- Enabled SSH
- Assigned hostname
- Connected via SSH
- Verified internet connectivity

Software

- Raspberry Pi OS Lite
- Raspberry Pi Imager
- Windows Powershell

Commands Used

`</> Bash`

`ssh student@192.168.x.x`

hostname

whoami

ip addr

ip route

ping google.com

Verification/Lessons Learned

- Successfully connected through SSH.
- Internet connectivity verified.
- IP address confirmed.
- Hostname verified.
- Learned how SSH established a secure remote connection.
- Learned how to view interface configuration.
- Learned basic Linux navigation.

Network Diagram

Internet

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Router (192.168.x.x)

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Windows PC (192.168.x.x) Raspberry Pi (192.168.1.246)

Screenshots

```
student@pi-lab:~$  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
student@pi-lab:~$ hostname  
pi-lab  
student@pi-lab:~$ whoami  
student  
student@pi-lab:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether b8:27:eb:14:71:23 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.1.1/24 brd 192.168.1.255 scope global dynamic noprefixroute wlan0  
        valid_lft 8551sec preferred_lft 8551sec  
    inet6 2001:4840:1c00:1000::b827:eb14:7123/64 scope global dynamic mngtppaddn proto kernel ra  
        valid_lft 2722sec preferred_lft 2722sec  
    inet6 fe80::b827:eb14:7123/64 scope link proto kernel ll  
        valid_lft forever preferred_lft forever  
student@pi-lab:~$ ip route  
default via 192.168.1.1 dev wlan0 proto dhcp src 192.168.1.240 metric 600  
0: * 1: 1: 0: 2: 4 dev wlan0 proto kernel scope link src 192.168.1.240 metric 600  
student@pi-lab:~$ ping -c 4 google.com  
ping: invalid argument: google.com  
student@pi-lab:~$ ping -c 4 google.com  
PING google.com (2607:f8b0:4000:804::200e) 56 data bytes  
64 bytes from iclga-at-in-xbe.icl80.net (2607:f8b0:4000:804::200e): icmp_seq=1 ttl=117 time=5.87 ms  
64 bytes from iclga-at-in-xbe.icl80.net (2607:f8b0:4000:804::200e): icmp_seq=2 ttl=117 time=9.45 ms  
64 bytes from iclga-at-in-xbe.icl80.net (2607:f8b0:4000:804::200e): icmp_seq=3 ttl=117 time=14.1 ms  
64 bytes from iclga-at-in-xbe.icl80.net (2607:f8b0:4000:804::200e): icmp_seq=4 ttl=117 time=53.4 ms  
--- google.com ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3000ms  
rtt min/avg/max/mdev = 5.878/20.609/53.389/10.408 ms  
student@pi-lab:~$ time ls  
real    0m0.026s  
user    0m0.016s  
sys     0m0.015s  
student@pi-lab:~$ time hostname  
real    0m0.016s  
user    0m0.011s  
sys     0m0.003s  
student@pi-lab:~$ cat /etc/resolv.conf  
# Generated by NetworkManager  
search mynetworksettings.com  
nameserver 192.168.1.1  
student@pi-lab:~$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=8.93 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=11.6 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=12.8 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=15.5 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=118 time=8.22 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=118 time=12.1 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=118 time=15.2 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=118 time=56.9 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=118 time=17.9 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=118 time=12.4 ms
```

```
student@pi-lab:~$  
# Generated by NetworkManager  
search mynetworksettings.com  
nameserver 192.168.1.1  
student@pi-lab:~$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
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64 bytes from 8.8.8.8: icmp_seq=8 ttl=118 time=56.9 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=118 time=17.9 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=118 time=12.4 ms  
64 bytes from 8.8.8.8: icmp_seq=11 ttl=118 time=9.22 ms  
64 bytes from 8.8.8.8: icmp_seq=12 ttl=118 time=10.2 ms  
64 bytes from 8.8.8.8: icmp_seq=13 ttl=118 time=12.9 ms  
64 bytes from 8.8.8.8: icmp_seq=14 ttl=118 time=11.1 ms  
64 bytes from 8.8.8.8: icmp_seq=15 ttl=118 time=8.11 ms  
64 bytes from 8.8.8.8: icmp_seq=16 ttl=118 time=71.3 ms  
64 bytes from 8.8.8.8: icmp_seq=17 ttl=118 time=81.6 ms  
64 bytes from 8.8.8.8: icmp_seq=18 ttl=118 time=24.4 ms  
64 bytes from 8.8.8.8: icmp_seq=19 ttl=118 time=8.78 ms  
64 bytes from 8.8.8.8: icmp_seq=20 ttl=118 time=82.8 ms  
64 bytes from 8.8.8.8: icmp_seq=21 ttl=118 time=12.9 ms  
64 bytes from 8.8.8.8: icmp_seq=22 ttl=118 time=10.7 ms  
64 bytes from 8.8.8.8: icmp_seq=23 ttl=118 time=58.6 ms  
64 bytes from 8.8.8.8: icmp_seq=24 ttl=118 time=11.0 ms  
^C  
--- 8.8.8.8 ping statistics ---  
24 packets transmitted, 24 received, 0% packet loss, time 23037ms  
rtt min/avg/max/mdev = 8.107/23.217/81.620/22.808 ms  
student@pi-lab:~$
```